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Likelihood-based inference for generalized linear mixed models: Inference with the R package glmm

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The R package glmm enables likelihood-based inference for generalized linear mixed models with a canonical link. No other publicly available software accurately conducts likelihood-based inference for generalized linear mixed models with crossed random effects. glmm is able to do so by approximating the likelihood function and two derivatives using importance sampling. The importance sampling distribution is an essential piece of Monte Carlo likelihood approximation, and developing a good one is the main challenge in implementing it. The package glmm uses the data to tailor the importance sampling distribution and is constructed to ensure finite Monte Carlo standard errors. In the context of the generalized linear mixed model, the salamander model with crossed random effects has become a benchmark example. We use this model to illustrate the complexities of the likelihood function and to demonstrate the use of the R package glmm.

KEYWORDS
computationally intensive methods, generalized linear models, likelihood, regression, statistical inference, statistical modelling

1 | INTRODUCTION

The generalized linear mixed model (GLMM) extends the generalized linear model (GLM) by allowing both fixed effects and mean-zero normal random effects (Stiratelli, Laird, & Ware, 1984). Despite the widespread use of GLMs in applications, the use of GLMMs has been limited since—except for some special cases—the methods of frequentist likelihood-based inference were not available in software until the recent release of the R package glmm (Knudson, Geyer, & Benson, 2020). GLMMs are computationally difficult because the likelihood function involves an intractable integral that is often complicated or high-dimensional in applications.

Other software uses a variety of approaches to dealing with the intractable likelihood. The SAS procedure NLMIXED uses adaptive Gaussian quadrature to approximate the integral, which limits its applicability to independent random effects of small dimension which are not crossed ("crossed" random effects are defined in Section 2.1). The R package lme4 (Bates, Mächler, Bolker, & Walker, 2015) avoids the likelihood and uses a Laplace approximation or adaptive Gauss–Hermite quadrature for models limited to a single scalar random effect. The SAS procedure GLIMMIX defaults to either penalized quasi-likelihood (PQL) or Laplace approximations, but adaptive Gaussian quadrature is available for a limited class of models. The R package MASS (Venables & Ripley, 2002) uses PQL. The bernor package (Sung & Geyer, 2007) uses a Monte Carlo likelihood approximation (MCLA) to the likelihood function (Geyer, 1994; Thompson & Guo, 1991; Geyer & Thompson, 1992) but cannot accurately perform maximum likelihood for the benchmark salamander data with crossed random effects; see Section 2.2.

With the release of the R package glmm, frequentist likelihood-based inference is available in Binomial or Poisson GLMMs, including calculating Fisher information, hypothesis testing and constructing likelihood-based confidence intervals. It is the only publicly available software that accurately conducts likelihood-based inference for GLMMs for both nested and crossed random effects. The glmm package uses MCLA to approximate the likelihood function. MCLA is an instance of importance sampling, and the essential innovation that allowed effective use of MCLA in glmm is the development of a novel importance sampling distribution.

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2 | GENERALIZED LINEAR MIXED MODELS

2.1 | Model description

In a GLM, the linear predictor η is related linearly to the coefficients vector β as $\eta = X\beta$, where X is the design matrix and η is related to the mean vector μ by the link function, which is a coordinate-wise monotone transformation. In a GLMM, none of this changes, but the linear predictor also includes random effects, which are components of the random effects vector U. Thus, now, $\eta = X\beta + ZU$, where X is the design matrix for fixed effects and Z is the design matrix for random effects. If Y is the response vector, $\mu = E(Y|U = u)$. We consider only canonical link functions: logit for the binomial family and log for the Poisson family.

The random effects are assumed normal with mean zero. In some GLMM implementations, random effects are allowed to be dependent with a few forms of dependence, but the `R` package `glmm` assumes the random effects are independent. Components of the vector ν are variances of components of U. That is, the variance matrix for U is diagonal, and the diagonal entries are the components of ν. Some forms of dependent random effects can be accommodated by redefining the random effects, as in section 3.4 of Sung and Geyer (2007). Crossed random effects can present a particular challenge for fitting GLMMs because the likelihood may be unsuitable for evaluation by numerical integration; see Section 2.3. To see what we mean by 'crossed,' consider a graph with a node for every random effect. If a component of the response vector depends on two random effects, those random effects' nodes are connected with an edge. If each maximal connected subgraph contains multiple random effects from multiple variance components, then the random effects are crossed.

2.2 | The salamander example

To illustrate the package `glmm`, we introduce the salamander example. The salamander data and model have become a benchmark; (McCullagh & Nelder, 1989, section 14.5) first presented the experiment and data, Karim and Zeger (1992) proposed a model they call "Model A," and numerous researchers have reanalyzed the data (see, among others, Booth & Hobert, 1999; Breslow & Clayton, 1993; Karim & Zeger, 1992; Schall, 1991; Sung & Geyer, 2007; Wolfinger & O'Connell, 1993). The data set is included in the `glmm` package under the name `salamander`.

2.2.1 | The salamander experiment

Female salamanders and male salamanders of a single species were collected from two locations. Based on their location of origin, the salamanders were categorized into populations named 'Rough Butt' and 'White Side.' The scientific goal was to determine whether salamanders were more likely to mate with those from their own population or whether they were equally likely to mate with salamanders from either population. More specifically, scientists sought the odds of mating for each type of cross: female Rough Butt salamanders and male Rough Butt salamanders (denoted R/R), female Rough Butt salamanders and male White Side salamanders (denoted R/W), female White Side salamanders and male Rough Butt salamanders (denoted W/R) and, female White Side salamanders and male White Side salamanders (denoted W/W).

The experiment was conducted three times. Each experiment consisted of trials conducted on two closed groups of 20 salamanders. That is, each experiment was conducted on 40 salamanders that were split into two closed groups. Each group contained five female Rough Butts, five female White Sides, five male Rough Butts, and five male White Sides. Each trial consisted of placing a female salamander and a male salamander in an isolated space together, then observing whether the salamanders mated. Each female salamander participated in six trials with male salamanders from her closed group: three trials with male White Side salamanders and three trials with male Rough Butt salamanders. Scientists paired salamanders from the same group; intergroup trials were not conducted. Thus, 60 trials were conducted on each closed group, each experiment consisted of 120 trials, and the overall data set contains binary responses from 360 trials. Although the same salamanders were used in the first two experiments, the data are traditionally modelled assuming that new salamanders were used in each experiment (Booth & Hobert, 1999; Karim & Zeger, 1992; McCullagh & Nelder, 1989). Due to the experimental design, the males' and females' random effects are crossed. A graph (as described in Section 2.1) for this example would have six maximal connected subgraphs, and each maximal connected subgraph would have 20 random effect nodes.

2.2.2 | The salamander data set

The following two `R` commands provide access to the `glmm` package and the `salamander` data frame.

```
> library(glmm)
> data(salamander)
```

To familiarize the reader with the data, we display a few rows of the data and the class of each variable.

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```
> salamander[5:7, ]
  Mate  Cross  Female  Male
5    1    R/R      14    12
6    1    R/W      15    28
7    0    R/W      16    27
> sapply(salamander, class)
     Mate    Cross    Female    Male
"numeric" "factor" "factor" "factor"
```

The variable `Mate` indicates whether the pair of salamanders mated: The value is 1 if they mated and 0 if they did not. The factor `Cross` indicates the female's type and then the male's type, as described in Section 2.2.1. The factors `Female` and `Male` contain identification numbers for the female salamander and male salamander, respectively.

2.2.3 | The salamander model

Model A (Karim & Zeger, 1992) includes a random effect for each female salamander; a random effect for each male salamander; a fixed effect predictor for each of the four types of cross; and a Bernoulli response of whether the pair of salamanders mated, dependent on the type of cross, the female random effect and the male random effect. Each salamander's random effect is assumed to be independent of the others. All random effects are assumed to be normally distributed with mean 0. The male salamanders' random effects share one variance component (denoted v_M) while the female salamanders' random effects share another (denoted v_F).

Let π_i represent the probability of successful mating for salamander pair i. Let $I()$ denote the function that maps false and true to zero and one, respectively, so $I(\text{Cross} = \text{R/R})$ is 1 when the variable Cross has the value R/R and 0 otherwise. Let u_i^F represent the random effect from the female salamander in the ith pair. Let u_i^M represent the random effect from the male salamander in the ith pair. Then the model can be written as follows.

$$\begin{aligned}\log\left(\frac{\pi_i}{1-\pi_i}\right) &= \beta_{\text{R/R}} I(\text{Cross} = \text{R/R}) + \beta_{\text{R/W}} I(\text{Cross} = \text{R/W}) + \beta_{\text{W/R}} I(\text{Cross} = \text{W/R}) + \beta_{\text{W/W}} I(\text{Cross} = \text{W/W}) + u_i^F + u_i^M \\ u_i^F &\overset{iid}{\sim} N(0, v_F) \\ u_i^M &\overset{iid}{\sim} N(0, v_M)\end{aligned} \tag{1}$$

2.3 | The likelihood function

The likelihood is a function of the parameters given the data. Multiplicative terms that do not contain parameters can be dropped from the likelihood. The linear predictor η depends on the random effects, but random effects cannot be part of the likelihood because they are not observed data. Let $f_\beta(y|u)$ denote the density for the data conditional on the random effects, $f_v(u)$ denote the density for the random effects and $f_\theta(u, y)$ denote the joint density for the random effects and the data. Then the likelihood for a GLMM is

$$L(\theta|y) = \int f_\beta(y|u) f_v(u)\, du = \int f_\theta(u, y)\, du. \tag{2}$$

Often, this integral cannot be expressed in closed form, but the random effects' structure may allow the likelihood to be factored into a product of lower-dimensional integrals. If each component of the response vector depends on a single random effect, then the likelihood can be factored into a product of one-dimensional integrals, making numerical integration (e.g., adaptive Gaussian quadrature) possible.

However, if the random effects are crossed, the integral in Equation (2) cannot be factored into a product of one-dimensional integrals. Rather, the likelihood can be factored into a product of integrals with dimension determined by the number of random effect nodes in the maximal connected subgraph. For example, the salamander likelihood can be factored into a product of six 20-dimensional integrals but no further. Numerical integration is often not possible for models with crossed random effects due to the dimension of the integral, forcing us to turn to alternative methods such as MCLA.

3 | MONTE CARLO LIKELIHOOD APPROXIMATION

MCLA produces a Monte Carlo approximation to the likelihood via importance sampling. Maximum likelihood is often of interest: The maximizer of the MCLA is called the Monte Carlo maximum likelihood estimator (MCMLE). Under a Wald-type integrability condition, the likelihood approximation converges almost surely to the likelihood function, and the MCMLE converges to the MLE almost surely as the Monte Carlo sample size increases (Geyer, 1994).

MCLA is applicable in a variety of settings (Geyer, 1990; Thompson & Guo, 1991; Gelfand & Carlin, 1993), but we focus on MCLA for normalized densities with random effects. Prior to `glmm`, Sung and Geyer (2007) prepared the `R` package `bernor` that fits maximum likelihood estimates for

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a GLMM with a Bernoulli response; their package uses an importance sampling distribution that does not depend on the observed data. Although their package can perform maximum likelihood for simpler models, it cannot find MLEs for a model as complicated as the salamander model (described in Section 2.2.3).

The limitations of `bernor` highlight the challenge of MCLA: finding an importance sampling distribution that performs well in practice. Although MCLA theory indicates any importance sampling distribution should suffice as long as its support contains the support of the target distribution $f_\theta(u, y)$, many importance sampling distributions require impracticably large Monte Carlo sample sizes.

3.1 | Calculation of the MCLA

MCLA requires an importance sampling distribution, which we denote $\tilde{f}(u)$. We can rewrite (2) as

$$L(\theta|y) = E_{\tilde{f}}\left\{\frac{f_\theta(u, y)}{\tilde{f}(u)}\right\}. \tag{3}$$

If $u_1, \ldots, u_m$ are realizations from $\tilde{f}$, then the MCLA is as follows:

$$L_m(\theta|y) = \frac{1}{m}\sum_{k=1}^{m}\frac{f_\theta(u_k, y)}{\tilde{f}(u_k)}. \tag{4}$$

The distribution $\tilde{f}(u)$ has some restrictions. It is not allowed to depend on θ because that would make derivatives of (3) hard to approximate, whereas they are approximated by derivatives of (4) if $\tilde{f}(u)$ does not depend on θ (Knudson, 2016, section 3.1). The distribution $\tilde{f}(u)$ is allowed to depend on y (and does depend on y in the implementation in the `R` package `glmm`). To avoid dividing by zero in (3) and (4), we require that $\tilde{f}(u) = 0$ implies $f_\theta(u, y) = 0$ for all θ and for the observed value of y. Stronger conditions are needed for almost sure convergence of estimators, and even stronger conditions are needed for asymptotic normality of estimators (Section 3.3 below).

3.2 | Implementation of MCLA in `glmm`

The importance sampling distribution used in `glmm` depends on the observed data and is a mixture of three distributions: two normal distributions and a T distribution. The T distribution is centred at 0 with five degrees of freedom and a covariance matrix based on the penalized quasi-likelihood (PQL) estimates of the variance components. The first normal distribution is centred at the PQL predictions of the random effects, and its covariance matrix is the PQL-estimated covariance matrix. The second normal distribution is centred at the PQL predictions of the random effects and has a covariance matrix based on the Hessian of the penalized likelihood from PQL. More specifically, the Hessian of the log density of the second normal distribution matches the Hessian of the log density of the target distribution $f_\theta(u, y)$. By default, the importance sampling distribution evenly weights the three distributions, and the T distribution has 5 degrees of freedom, but the user can alter the weights and the degrees of freedom.

3.3 | Asymptotic properties of the MCLA

The asymptotic properties of the likelihood approximation as the Monte Carlo sample size increases have been studied in generality and for the specific implementation in `glmm`. Geyer (1994) presents conditions under which the MCLA converges almost surely to the exact likelihood for any single parameter value. The MCLA converges almost surely to the likelihood for GLMMs specified with unnormalized densities as long as a Wald-like integrability condition is met (Geyer, 1994). Additionally, if the parameter space can be compactified, the MCMLE converges to the MLE almost surely (Geyer, 1994). Geyer (1994) also shows that the Monte Carlo profile likelihoods converge to the exact profile likelihoods almost surely, and no additional regularity conditions are required. The importance sampling distribution implemented in `glmm` ensures that the Monte Carlo standard errors of the MCMLEs are finite (Knudson, 2016). This result depends on specific details of the importance sampling distribution (in particular, the inclusion of the T distribution), which were chosen with this goal in mind.

4 | USING THE R PACKAGE GLMM

4.1 | Formatting the data

To fit GLMMs using the `glmm` package, categorize the data into three types of vectors:

1. One to two vectors that contain the response data.
   (a) If the response is Poisson, then components of the single response vector must be nonnegative integers.
   (b) If the response is Bernoulli, then components of the single response vector must be zero or one.

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(c) If the response is binomial (with some sample sizes exceeding one), then (as with the R function `glm`) the response is a two-column matrix with the first column giving the numbers of successes and the second column giving the numbers of failures.

2. At least one vector that will be used for defining the random effects' design matrix. These should be of class `factor`.
3. Vectors that will be used for defining the fixed effects design matrix. These can be of class `factor` or `numeric`.

The R output in Section 2.2.2 confirms the `salamander` data meet the formatting guidelines: The data set has the response vector (`Mate`), a vector that defines the fixed effects (`Cross`) and two vectors that define the random effects (`Female` and `Male`).

4.2 | Using the model-fitting function

Before we fit the model, we create a cluster of four cores using `makeCluster` from the `parallel` package, and we set the seed to ensure reproducibility.

```
> set.seed(1234)
> clust <- makeCluster(4)
```

The code below fits the salamander model described in Section 2.2.3 and saves the result as `sal`.

```
> sal <- glmm(Mate~0+ Cross, random=list(~0+Female,~0+Male), varcomps.names=c("F","M"), data=salamander,
+  family.glmm=bernoulli.glmm, m=10^6, cluster=clust)
```

The fixed and random effects specifications both use the R formula mini-language. The first formula specifies the fixed effects. The next two formulas—which are combined in a list as argument `random`—specify the random effects. Because `Mate` is the response, it is on the left of the tilde operator. We only need to specify the response in the first formula. The formula `Mate ~ 0 + Cross` calculates the log odds for each of the four levels of `Cross`. Alternatively, the formula `Mate ~ Cross` would specify the same statistical model with a different parameterization: In the formula mini-language, omitting `0 +` creates a model with a reference group.

The next two formulas `~ 0 + Female` and `~ 0 + Male` create a random effect for each female and a random effect for each male. We almost always want to include `0 +` in the random effects specification because we almost never want a reference group in the random effects. In other words, omitting `0 +` creates a random effect structure with a different (likely illogical) scientific interpretation.

Following the `random` list, the argument `varcomps.names` assigns names to the variance components. Because the `random` list placed the females first, the order of the variance components names is first `~F~` and then `~M~`. The last argument of the first line shows we have the option to specify the data set; if the data set is not specified, `glmm` looks to the parent environment for the variables referenced.

The second line shows we must specify the distribution of the response. In the salamander mating example, the response is binary (the salamanders either mated or they did not), and we therefore state `bernoulli.glmm`, although the more general `binomial.glmm` would specify the same model. The family for a Poisson response is `poisson.glmm`.

Next, we specify the Monte Carlo sample size `m`. A larger Monte Carlo sample size produces a more accurate MCLA, which, in turn, produces more accurate Monte Carlo maximum likelihood estimates (MCMLEs). We want the largest `m` that time allows. We can check the Monte Carlo standard errors (as shown in Section 4.5) to decide whether a particular Monte Carlo sample size is large enough for our needs. Finally, we have the option to pass in a cluster in order to complete the MCLA computations in parallel. This example creates and uses a cluster with four cores. More information on parallel computing can be found in Section 6.

Additional arguments may be added to incorporate relevance weights, alter the mixture weights of the importance sampling distribution from the default of even thirds, set variance components equal, adjust optimization arguments, or output additional information. These options are intended for advanced users and are described in the package documentation, which can be found with `help(glmm)`.

4.3 | Reading the summary output

After fitting the model, we use the `summary` command, which displays

- the function call;
- the link function;
- the fixed effect estimates, their standard errors (calculated using observed Fisher information), their `z value` test statistics (testing whether the coefficients are significantly different from zero), the test's p values, and the R-standard significance stars (optional);
- the variance component estimates, their standard errors (calculated using observed Fisher information), their `z value` test statistics (testing whether the coefficients are significantly different from zero), the test's p values, and the R-standard significance stars (optional).

The p value for the fixed effects is calculated using a two-sided alternative hypothesis ($H_A : \beta \neq 0$) while the p value for the variance components is calculated using a one-sided alternative hypothesis ($H_A : \nu > 0$) because variance components must be nonnegative.

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```
> summary(sal)
Call:
glmm(fixed = Mate ~ 0 + Cross, random = list(~0 + Female, ~0 + Male), varcomps.names = c("F", "M"),
     data = salamander, family.glmm = bernoulli.glmm, m = 10^6, cluster = clust)


Link is: "logit (log odds)"
Fixed Effects:
         Estimate Std. Error z value Pr(>|z|)
CrossR/R    1.000      0.384   2.600  0.00931 **
CrossR/W    0.320      0.390   0.823  0.41029
CrossW/R   -1.940      0.450  -4.263 2.02e-05 ***
CrossW/W    1.010      0.430   2.333  0.01965 *
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Variance Components for Random Effects (P-values are one-tailed):
  Estimate Std. Error z value Pr(>|z|)/2
F     1.34       0.52   2.597     0.0047 **
M     1.24       0.56   2.190     0.0143 *
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The summary provides the estimates of the parameters in Equation (1)

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = 1.000\,I(\text{Cross} = \text{R/R}) + 0.320\,I(\text{Cross} = \text{R/W}) - 1.940\,I(\text{Cross} = \text{W/R}) + 1.010\,I(\text{Cross} = \text{W/W}) + u_i^F + u_i^M$$
$$u_i^F \overset{iid}{\sim} N(0, \hat{v}_F = 1.34)$$
$$u_i^M \overset{iid}{\sim} N(0, \hat{v}_M = 1.24).$$

4.4 | Estimating the variance–covariance matrix

The variance–covariance matrix for the parameter estimates can be found using the `vcov` function. The only input is the model name.

```
> print(myvcov <- vcov(sal))
           CrossR/R     CrossR/W     CrossW/R   CrossW/W           F           M
CrossR/R 0.14778548  0.043112283  0.029517453 0.01160385  0.01718023  0.01597265
CrossR/W 0.04311228  0.155659297 -0.003403946 0.05754907  0.02769045  0.01078458
CrossW/R 0.02951745 -0.003403946  0.206389801 0.02876690 -0.05257510 -0.06082939
CrossW/W 0.01160385  0.057549073  0.028766896 0.18868124  0.05671537  0.04916507
F        0.01718023  0.027690446 -0.052575102 0.05671537  0.26799523  0.05036898
M        0.01597265  0.010784583 -0.060829393 0.04916507  0.05036898  0.31919849
```

The variance–covariance matrix can be useful for hypothesis testing. For example, to answer the original research question ('Are the log odds of mating different among the four crosses?'), we use a Wald test with the following test statistic:

$$\left(\begin{bmatrix}1&-1&0&0\\1&0&-1&0\\1&0&0&-1\end{bmatrix}\begin{bmatrix}\hat{\beta}_{\text{R/R}}\\\hat{\beta}_{\text{R/W}}\\\hat{\beta}_{\text{W/R}}\\\hat{\beta}_{\text{W/W}}\end{bmatrix}\right)^T\left(\begin{bmatrix}1&-1&0&0\\1&0&-1&0\\1&0&0&-1\end{bmatrix}J\begin{bmatrix}1&-1&0&0\\1&0&-1&0\\1&0&0&-1\end{bmatrix}^T\right)^{-1}\left(\begin{bmatrix}1&-1&0&0\\1&0&-1&0\\1&0&0&-1\end{bmatrix}\begin{bmatrix}\hat{\beta}_{\text{R/R}}\\\hat{\beta}_{\text{R/W}}\\\hat{\beta}_{\text{W/R}}\\\hat{\beta}_{\text{W/W}}\end{bmatrix}\right) \tag{5}$$

where J is the variance–covariance matrix for $\hat{\beta}_{\text{R/R}}, \hat{\beta}_{\text{R/W}}, \hat{\beta}_{\text{W/R}}, \hat{\beta}_{\text{W/W}}$; that is, J is `myvcov[1:4, 1:4]`. This test statistic has a χ^2 distribution with 3 degrees of freedom. Executing this test in `R` produces a p value of 1.676×10^{-8}, indicating that we should retain the four fixed effects rather than reducing our model to a single fixed effect.

4.5 | Calculating standard errors and Monte Carlo standard errors

With maximum likelihood performed by MCLA, there are two sources of variability: There is variability from sample (of observed data) to sample and from Monte Carlo sample (of generated random effects) to Monte Carlo sample. The first source of variability (from sample to sample) is

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	$\hat{\beta}_{R/R}$	$\hat{\beta}_{R/W}$	$\hat{\beta}_{W/R}$	$\hat{\beta}_{W/W}$	$\hat{\nu}_F$	$\hat{\nu}_M$
Knudson et al. (2020) (glmm)	1.00	0.32	−1.94	1.01	1.34	1.24
Booth and Hobert (1999) (MCEM)	1.03	0.32	−1.95	0.99	1.40	1.25
Bates et al. (2015) (lme4)	1.01	0.31	−1.90	0.99	1.17	1.04

TABLE 1 Point estimates for salamander data

measured using the usual standard error. The second source of variability is assessed by the Monte Carlo standard error. The standard error decreases as the (observed data) sample size increases; similarly, the Monte Carlo standard error decreases as the Monte Carlo sample size (`m`) increases.

Standard errors for estimates appear with the point estimates in the `summary` tables and can also be found using the command `se`. The standard errors are calculated using the diagonal of the variance covariance matrix (`vcov`). The argument for `se` is the model.

```
> se(sal)
CrossR/R   CrossR/W  CrossW/R  CrossW/W         F         M
0.3844288 0.3945368 0.4543014 0.4343745 0.5176826 0.5649765
```

We can calculate the Monte Carlo standard errors with `mcse`. The argument for `mcse` is the model.

```
> mcse(sal)
   CrossR/R    CrossR/W    CrossW/R    CrossW/W           F           M
0.007364104 0.013587752 0.017257579 0.021866960 0.038427804 0.038940917
```

Users can inspect the Monte Carlo standard error to decide whether the Monte Carlo sample was large enough: Each Monte Carlo standard error should be small compared to its corresponding standard error so that the MCMLEs vary little when different random numbers are used in `glmm`.

4.6 | Additional functions

Additional `glmm` functionality includes fixed effect point estimation using the `coef` or `coefficients` functions, variance estimation using the `varcomps` command, and interval estimation using the `confint` command (which calls the `vcov` function).

5 | PERFORMANCE

Table 1 compares point estimates for Model A (Karim & Zeger, 1992) among `glmm` with a Monte Carlo sample size of $m = 10^6$, Monte Carlo EM (Booth & Hobert, 1999), and the Laplace-based `lme4` (Bates et al. 2015). The MCMLEs produced by `glmm` match the MCMLEs produced by Booth and Hobert (1999). Markov chain Monte Carlo (MCMC) confirms that these estimates are MLEs by plugging the putative MLEs into the likelihood, running an MCMC sampler, and checking that 0 is in the credible interval for the gradient of the likelihood. The fixed effect estimates from `lme4` match the `glmm` and Booth and Hobert (1999) results, but the `lme4` variance component estimates are a bit smaller than those found by `glmm` and Booth and Hobert (1999).

6 | NOTES ON PROGRAMMING MONTE CARLO METHODS

By definition, Monte Carlo methods rely on repeated random sampling. Both the randomness of the random sampling and the repetition require extra care from programmers relying on Monte Carlo methods. In this section, we explain the challenges of Monte Carlo and the steps we have taken as programmers to address these challenges.

First, the randomness inherent to Monte Carlo methods can make the methods difficult to check. To test `glmm`'s likelihood approximation and all other calculations dependent on the generated random effects, we removed the randomness by setting the seed used for random number generation and using identical sets of generated random effects. With an unchanging set of generated random effects and an unchanging set of observed data, the likelihood approximation (and all resulting inference) is subject to no randomness. Therefore, we have checked the likelihood approximation (and other calculations dependent on the generated random effects) by using identical sets of generated random effects.

Second, the repeated random sampling required for Monte Carlo methods can be computationally expensive. To save time and computing resources, we programmed the likelihood calculation, its gradient vector, and its Hessian matrix in C. We then profiled the code (using the `Rprof` function) to identify computationally expensive functions in `glmm`. This profiling demonstrated that a vast majority (over 90% in our examples) of the `glmm` computing time is spent on C calculations. To further reduce computing time, we have programmed the calculations to be done in parallel with cores provided by the user. As demonstrated in Section 4.2, users simply pass a cluster to the model fitting function. For example, `glmm` fits Model A with a Monte Carlo sample size of 10^4 in 92 seconds with four cores and 226 seconds with a single core.

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7 | DISCUSSION

The package `glmm` performs and enables frequentist likelihood-based inference for GLMMs by implementing MCLA. Currently, `glmm` fits models for binomial and Poisson responses and incorporates independent random effects, which can be crossed. With the user-provided model and data, `glmm` approximates the likelihood function and performs maximum likelihood to find point estimates for fixed effects and variance components; calculates the variance-covariance matrix and standard errors of the resulting point estimates (with function `vcov`); constructs confidence intervals (with function `confint`); calculates the Monte Carlo standard error for the resulting point estimates (with function `mcse`); and evaluates the log likelihood, gradient of the log likelihood and Hessian of the log likelihood at the point estimates. The `glmm` output enables users to perform any method of frequentist likelihood-based inference.

The characteristic that differentiates `glmm` from other software for GLMMs is the method implemented: MCLA with a carefully constructed importance sampling distribution. Moreover, the importance sampling distribution in `glmm` is constructed with both statistical practice and statistical theory in mind. It is designed to use the observed data to mimic the target distribution, and this choice makes `glmm` successful where other MCLA implementations (such as the `R` package `bernor`) have failed. Additionally, the `glmm` importance sampling distribution is constructed to ensure finite Monte Carlo standard errors.

In the examples in Section 5, `glmm` produces point estimates that agree with those produced via Monte Carlo EM and `lme4` ; the advantage of `glmm` is it enables all likelihood-based inference while Monte Carlo EM is limited to maximum likelihood, and `lme4`'s function `glmer` does not perform likelihood-based inference.

Future development of this package includes expanding the response types, programming additional methods of likelihood-based inference, and expanding the random effects' structure.

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